Connecting Power BI to Claude via MCP Server

A setup guide for the AI-Assisted E-commerce Return Analysis project

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Why this guide exists

This is the setup process I used to connect Claude Desktop to Power BI Desktop using the official Microsoft Power BI Modeling MCP server, as part of my **AI-Assisted E-commerce Return Analysis** project (Excel → MCP server → Power BI → Claude). I'm documenting it step by step so it's easy to reproduce, and so I can link it directly from the project's GitHub repo:

```
github.com/jitendera-code/AI-Assisted-E-commerce-Return-Analysis-Excel-MCP-server-Power-BI-claude
```

What MCP actually does here

MCP (Model Context Protocol) is the connector that lets Claude talk directly to a running Power BI Desktop file instead of me copy-pasting DAX back and forth. Once it's wired up, I can ask Claude in plain English to create measures, build relationships, or inspect the data model, and it executes inside the .pbix file directly.

What Claude can do through MCP

Task	Manual effort saved
Read the full data model (tables, columns, types)	No clicking through each table
Write DAX measures from a plain-English request	No syntax lookup / debugging
Create several measures in one go	Minutes instead of an hour for a return-rate measure set
Build relationships between tables	No manual drag-and-drop in Model View
Create calculated tables (e.g. a Date table)	No hand-typed DAX for date logic
Run DAX queries and return results	Instant formula testing
Flag and help fix broken measures	Faster debugging

What still has to be done manually in Power BI

Still manual	Why
Importing the Excel return-data files	MCP can't touch the file system / Get Data
Power Query cleaning (types, blanks, renames)	MCP doesn't reach the Power Query engine
Building charts and visuals	MCP can't modify the report canvas
Formatting, colours, titles, slicers	MCP has no access to the visual layer
Publishing the report	Needs my Microsoft account sign-in

What you need before starting

- **Power BI Desktop** — already installed, with the return-analysis .pbix file ready.
- **Claude Desktop** — download from claude.ai/download if not already installed.
- **VS Code** — download from code.visualstudio.com. It's only used to install the MCP server extension; no coding required inside it.

Step 1 — Install the Power BI Modeling MCP extension in VS Code

- Open VS Code.
- Press **Ctrl+Shift+X** to open the Extensions panel.
- Search for **Power BI Modeling MCP**.
- Install the extension published by **Analysis Services** (Microsoft's official listing).
- Once it shows Installed, VS Code has done its job — it can be minimised.

Step 2 — Locate the MCP server executable

The extension drops a small executable on disk. Open a terminal in VS Code (Ctrl+`) and run:

```
dir "$env:USERPROFILE\.vscode\extensions\analysis-services*"
```

This prints a folder name similar to `analysis-services.powerbi-modeling-mcp-0.4.0-win32-x64`. The full path to the executable will be:

```
C:\Users\YOUR_USERNAME\.vscode\extensions\THE_FOLDER_NAME\server\powerbi-modeling-mcp.exe
```

Your Windows username is visible in File Explorer's address bar as C:\Users\YourName.

Step 3 — Point Claude Desktop at the MCP server

- Open Claude Desktop → Settings (gear icon) → Developer tab.
- Click **Edit Config** — this opens claude_desktop_config.json.
- Add (don't delete any existing config) an mcpServers entry, using the exact path from Step 2:

```
{
  "mcpServers": {
    "powerbi-modeling-mcp": {
      "command": "C:\\Users\\YOUR_USERNAME\\.vscode\\extensions\\analysis-services.powerbi
-modeling-mcp-0.4.0-win32-x64\\server\\powerbi-modeling-mcp.exe",
      "args": ["--start"],
      "type": "stdio"
    }
  }
}
```

Every backslash in the path must be doubled (\\) — this is a JSON escaping rule, not optional.

Step 4 — Fully restart Claude Desktop

Closing the window isn't enough — Claude keeps a background process running.

- **Preferred:** right-click the Claude icon in the system tray (near the clock) → Quit.
- **Fallback:** Ctrl+Shift+Esc → find every Claude process → End Task on each.
- Reopen Claude Desktop from the Start Menu.

Step 5 — Confirm the connection

Look at the chat input box in Claude Desktop. A small hammer/tools icon appearing there means the MCP server is live. No icon means something in the config is off — recheck the path and backslashes.

Verifying it works with your Power BI file

Before testing, open Power BI Desktop and load the return-analysis .pbix file — Claude connects to a running instance, not to a file sitting closed on disk.

In Claude Desktop, try:

```
Connect to my Power BI Desktop model
```

```
Show me all tables in my model with their column names
```

```
How many rows does each table have?
```

```
Are there any existing relationships in my model?
```

```
Are there any existing measures?
```

Troubleshooting

Problem	Fix
No hammer icon after restart	Re-open config.json; check path, double backslashes, matching braces
"Server disconnected"	Power BI Desktop must already be open with the .pbix loaded
"Cannot find path" error	Re-run the dir command; the folder name may differ from what you noted
Config already has other entries	Add mcpServers alongside them, separated by a comma — don't delete existing key
Stopped working after a Claude update	Updates can reset the config file; re-paste it and restart

How this fits my return-analysis workflow

- Import the 8 raw return-data Excel exports manually (Get Data → Excel).
- Clean types/blanks in Power Query manually.
- Connect Claude via MCP.
- Ask Claude to build the Date table, relationships, and the return-rate / refund measures.
- Build and format the actual visuals manually in Power BI.
- Use Claude to debug any measure that returns blank or errors.
- Publish manually.

Safety notes

- Save the .pbix before letting Claude make changes, so there's a fallback copy.
- Check Model View / Table View after each Claude change to confirm it applied correctly.
- Work on a copy of the file, not the live production report, while experimenting.
